AURAFIT AI

DOC 02 / 07

PRODUCT REQUIREMENTS DOCUMENT

What We're Building, and Why

Vision, target users, functional requirements, and success metrics for AuraFit AI v1.

Version **1.0**

September 2026

Confidential — Internal Use

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What's Inside

01 — WHY

Vision & Problem Statement

Most calorie and macro trackers require a cloud account, sync personal health data (weight, meal photos, body metrics) to a third-party server, and gate accurate food recognition behind a subscription. AuraFit AI takes the opposite position: **every gram of data and every model inference stays on the user's machine**, with zero recurring API cost and zero internet dependency for daily use.

The Privacy Gap

Health and body data is among the most sensitive information a person has. Cloud trackers turn it into a subscription asset; AuraFit AI treats it as the user's alone.

The Friction Gap

Manual food logging is the #1 reason people abandon tracking apps within two weeks. Point-and-shoot vision logging removes the search-and-select tedium.

The Cost Gap

Vision-AI food recognition today usually means a metered cloud API. A local Ollama runtime makes unlimited logging free after the one-time model download.

Vision statement: AuraFit AI is the calorie and body-transformation tracker for people who want serious, accurate tooling without handing their health data to anyone.

02 — WHO

Target Users

Rahul, 29 — Aggressive Fat Loss

PRIMARY PERSONA

CONTEXT

Software engineer, tracks macros in a spreadsheet today. Wants precision without another cloud subscription touching his health data.

NEEDS

- Fast photo-based logging between meetings
- Hard calorie ceiling with protein floor
- No account creation, no sync

SUCCESS LOOKS LIKE

Logs 90% of meals in under 20 seconds each, hits weekly deficit target consistently.

Meera, 24 — Lean Bulk, Vegetarian

SECONDARY PERSONA

CONTEXT

Competitive lifter following a vegetarian diet; struggles to hit protein targets and wants recipe ideas from what's already in her kitchen.

NEEDS

- Vegetarian-safe pantry recipe generation
- Surplus macro targets tuned to lean gain
- Clear allergy exclusion (lactose-sensitive)

SUCCESS LOOKS LIKE

Gets 3+ usable recipe suggestions a week that fit her remaining macro budget and pantry.

Daniel, 41 — Recomposition, Privacy-First

SECONDARY PERSONA

CONTEXT

Previously cancelled a popular tracking app after a data-breach headline. Wants long-term progress trends without any cloud footprint.

NEEDS

- Fully offline operation, verifiable via network monitor
- Local export of all historical data as CSV/JSON
- Long-horizon weight-velocity and goal-date charts

SUCCESS LOOKS LIKE

Trusts the app enough to log consistently for 6+ months; exports data for his own records annually.

03 — OUTCOMES

Goals & Success Metrics

<20s	0	±15%	100%
MEDIAN TIME TO LOG A MEAL PHOTO	NETWORK CALLS DURING CORE USE	TARGET CALORIE ESTIMATE ERROR BAND	DATA EXPORTABLE, USER-OWNED

Goal	Metric	Target
Reduce logging friction vs. manual search	Median seconds per logged meal	< 20s (vision path)
Maintain estimate trustworthiness	Mean absolute % error vs. golden-set ground truth	≤ 15%
Sustain daily engagement	7-day logging retention	≥ 60%
Prove privacy claim	Outbound network requests during core flows	0 (verified by automated smoke test)
Keep app lightweight	Idle RAM footprint (excluding Ollama)	< 40MB

04 — SCOPE

Functional Requirements

Requirements are grouped under the four core feature pillars. Priority: P0 required for v1 GA, P1 fast-follow, P2 future consideration.

4.1 Biometrics & Target Engine P0

ID	Requirement	Priority
BIO-01	Compute BMI, BMR (Mifflin-St Jeor), and TDEE from user-entered age, sex, height, weight, and activity level.	P0
BIO-02	Generate calorie targets and Protein/Carb/Fat/Fiber splits for goals: Aggressive Fat Loss, Lean Bulk, Recomposition, Maintenance.	P0
BIO-03	Accept and persist hard dietary constraints (Vegan, Vegetarian, Keto, Paleo, Halal, Kosher, Low-FODMAP) and allergy exclusions (Gluten, Lactose, Nuts, Shellfish).	P0
BIO-04	Recalculate targets automatically when weight, activity level, or goal changes.	P1

4.2 Local Vision AI Calorie Logger P0

ID	Requirement	Priority
VIS-01	Accept a meal photo via drag-and-drop or webcam capture.	P0
VIS-02	Send the image to a local llama3.2-vision instance and enforce structured JSON output (items, estimated weight, calories, protein, carbs, fat).	P0
VIS-03	Display an editable verification table before any entry is committed to the database.	P0
VIS-04	Fall back to manual/SQLite-search entry if the model output fails schema validation twice.	P0
VIS-05	Support multi-item plates (itemized breakdown, not a single aggregate guess).	P1

4.3 Smart Meal & Recipe Generator P1

ID	Requirement	Priority
REC-01	Generate meal suggestions that fit the user's remaining macro budget for the day.	P1
REC-02	Generate recipes from a user-entered pantry ingredient list, using a local text LLM (llama3.2 / qwen2.5).	P1
REC-03	Provide step-by-step instructions with measurements and prep time, fully offline.	P1
REC-04	Respect active dietary guardrails and allergy exclusions in every generated recipe.	P0

4.4 Local Database & Analytics Engine P0

ID	Requirement	Priority
DAT-01	Ship an embedded SQLite database pre-loaded with USDA nutrition data for zero-latency single-item lookup.	P0
DAT-02	Chart daily weight velocity, macro compliance, and logging history.	P0
DAT-03	Estimate a projected goal-completion date from recent weight trend.	P1
DAT-04	Export all user data to CSV and JSON on demand, with zero cloud sync.	P0

05 — VALIDATION

User Stories

US-01 · BIO-02

As Rahul, I want the app to compute my calorie and macro targets from my stats and goal, so that I don't have to research or calculate anything myself.

US-02 · VIS-01–03

As Rahul, I want to drop a photo of my lunch and get an editable macro breakdown in seconds, so that logging doesn't interrupt my day.

US-03 · REC-02–04

As Meera, I want to type in what's in my fridge and get a vegetarian, high-protein recipe, so that I don't default to the same three meals every week.

US-04 · DAT-04

As Daniel, I want a one-click export of every meal I've logged this year, so that my data is mine regardless of whether I keep using the app.

US-05 · VIS-04

As a new user, when the vision model returns something unusable, I want a clear manual fallback, so that I'm never blocked from logging a meal.

06 — RULES

Dietary Guardrail Matrix

Guardrails apply as hard filters across every surface that suggests food: the recipe generator, meal-plan recommendations, and even how the vision logger flags a conflicting item after logging.

Constraint Type	Options	Enforcement Point
Diet pattern	Vegan, Vegetarian, Keto, Paleo, Halal, Kosher, Low-FODMAP	Recipe/meal generation prompt constraints + post-generation filter
Allergy exclusion	Gluten, Lactose, Nuts, Shellfish	Hard exclusion list injected into every LLM prompt; flags matching ingredients on logged/generated items
Combination handling	Multiple constraints stack (e.g. Vegetarian + Lactose-free)	Constraints are ANDed; conflicting combinations surface a warning at onboarding

NOT A MEDICAL SAFETY SYSTEM

Guardrails reduce the chance of a diet-conflicting suggestion but are not a substitute for reading ingredient labels — this is stated explicitly in onboarding and the allergy settings screen.

07 — BOUNDARIES

Non-Goals & Constraints

Out of scope for v1

- Mobile companion app
- Multi-user / family accounts
- Wearable device integrations
- Barcode scanning (deferred to v1.1)
- Any cloud sync or account system

Constraints

- Vision inference requires a locally installed Ollama runtime
- llama3.2-vision:11b needs roughly 8GB+ available RAM
- USDA dataset licensing requires attribution in-app
- No telemetry beyond opt-in, anonymized crash reports